

In an operating system three CPU-intensive processes are ready for execution, which require 10ns, 20ns and 30ns and arrival at times 0ns, 2ns and 6ns, respectively. Write a program to calculate the total number of context switches needed if the operating system implements a shortest job first (preemptive) scheduling algorithm. Also calculate the average time for which the processes have to wait before getting the CPU.

CODE

```
main.c
1  #include <stdio.h>
2
3  int main() {
4      int arrival[3] = {0, 2, 6};
5      int burst[3] = {10, 20, 30};
6      int completion[3], waiting[3];
7      int i;
8
9      // Completion times from SJF preemptive
      scheduling
10     completion[0] = 10;
11     completion[1] = 30;
12     completion[2] = 60;
13
14     float avg_wait = 0;
15
16     for(i = 0; i < 3; i++) {
17         waiting[i] = completion[i] - arrival[i] -
            burst[i];
18         avg_wait += waiting[i];
19         printf("Waiting time of P%d = %d ns\n", i+1,
            waiting[i]);
    }
```

```
main.c
10     completion[0] = 10;
11     completion[1] = 30;
12     completion[2] = 60;
13
14     float avg_wait = 0;
15
16     for(i = 0; i < 3; i++) {
17         waiting[i] = completion[i] - arrival[i] -
            burst[i];
18         avg_wait += waiting[i];
19         printf("Waiting time of P%d = %d ns\n", i+1,
            waiting[i]);
20     }
21
22     avg_wait = avg_wait / 3;
23
24     printf("\nTotal Context Switches = 2");
25     printf("\nAverage Waiting Time = %.2f ns",
        avg_wait);
26
27     return 0;
28 }
```

OUTPUT

Output

Clear

Waiting time of P1 = 0 ns
Waiting time of P2 = 8 ns
Waiting time of P3 = 24 ns

Total Context Switches = 2
Average Waiting Time = 10.67 ns

=== Code Execution Successful ===